

Practical work Serie 1

Exercise 1

Write a **C program** that calculates and displays the absolute value of the difference between two real numbers entered from the keyboard, without using the predefined absolute value function.

Exercise 2

Write a **C program** that allows you to sort and display two integer numbers (A and B) entered via the keyboard in ascending order ($A \leq B$); A should contain the smallest value regardless of its initial value.

Exercise 3

Write a **C program** that asks the user for a child's age (in years). Then he informs him of his sporting category (using the **switch** instruction):

- "Chicken" : from 6 to 7 years
- "Pupil" : from 8 to 9 years
- "Minimum" : from 10 to 11 years
- "Younger" : from 12 to 14 years
- "Junior" : from 15 to 17 years

Supplementary series of Practical Work

Exercise 1

We have three integers A , B , and C .

Write a **C program** that transfers the value of B to A , the value of C to B , and the value of A to C .

Exercise 2

Write a **C program** that requests three grades and their coefficients, then calculates and displays their average.

Exercise 3

Write a **C program** that reads two numbers m and n from the user and informs whether their product (without calculating the product) is negative, positive, or zero.

Exercise 4

Write a **C program** that asks the user for the width and length of a rectangle (in meters), then calculates and displays the perimeter and the area of the rectangle.

Exercise 5

Write a **C program** that calculates and displays the volume of a sphere.

Exercise 6

Write a **C program** that reads two real numbers (X and Y) and:

- Calculates and displays the sum of X and Y if they are positive.
- Calculates and displays the product of X and Y if they are not positive.
- Calculates and displays the difference between these two numbers.

Exercise 7

Write a **C program** that sorts and displays three integers (A , B , and C) read from the keyboard in descending order ($A \geq B \geq C$); A contains the highest value, and C contains the smallest.

Exercise 8

Let X be a strictly positive integer.

- Write a **C program** that reads the value of X and checks if X is divisible by both 5 and 7 without using logical operators.
- Write a **C program** that reads the value of X and checks if X is divisible by both 5 and 7 using logical operators.
- Write a **C program** that reads the value of X and checks if X is divisible by either 5 or 7 without using logical operators.
- Write a **C program** that reads the value of X and checks if X is divisible by either 5 or 7 using logical operators.

Exercise 9

Write a **C program** that reads a real number and:

- Checks and displays whether the number is positive, negative, or zero.
- Checks and displays whether the number is even or odd.
- Displays its quarter, its third, and its half of the number.

Exercise 10

Write a **C program** that displays the type of a letter (vowel or consonant) entered by the user using the **switch** statement.

Exercise 11

Write a **C program** that allows the user to enter three real numbers, then based on a choice from a menu displayed on the screen, calculates and displays the sum, product, average, finds the minimum, or finds the maximum of the three numbers.

The menu should appear on the screen as follows:

```
----- MENU -----
1 ----- Sum -----
2 ----- Product -----
3 ----- Average -----
4 ----- Minimum -----
5 ----- Maximum -----
Enter your choice:
```

Exercise 12

Write a **C program** for hotel pricing. The rates are as follows:

- 7500 Dinars for an adult in a single room.
- 6000 Dinars per room for three or more single rooms.
- 9500 Dinars for two adults in a double room.
- 8000 Dinars per room for four or more double rooms.
- Free for the first child.
- 3000 Dinars per child from the second child onwards.

Breakfast is mandatory and costs 300 Dinars per person, regardless of age. A double room must accommodate exactly two people.

To calculate the total price, the following information must be provided: the number of single rooms, double rooms, children, and nights.

Exercise 13

A classroom contains 200 seats arranged in 10 rows and 20 columns. Each student has a number between 1 and 200. During an exam, students are seated according to their numbers starting from the first row, then the second row, and so on, as shown in the following table:

	Column 1	Column 2	...	Column 20
Row 1	1	2	...	20
Row 2	21	22	...	40
...	...	...	...	...
Row 10	181	182	...	200

Write a **C program** that takes an integer between 1 and 200 as input and displays the row and column where the student should sit.

For example, if $n = 35$, the student should sit in the 2nd row and the 15th column.

Exercise 14

Write a **C program** that asks the user to enter a character, then displays whether it is an uppercase or lowercase letter. If it is not a letter, it returns an error message